

Syndrome Extraction Circuits with Near-Optimal Depths for Practical Quantum Error Correcting Code Families

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Abstract

Fault-tolerant quantum computing relies on low-depth and high-fidelity syndrome extraction for quantum error correcting codes. We propose a scheduling strategy for syndrome extraction circuits for the broad class of quasi-abelian lifted-product codes, encompassing both hypergraph product and bicycle codes. Our approach constructs syndrome extraction circuits with CNOT depths no more than one layer above the fundamental lower bound, frequently achieving optimal depth. A pipelined variant further reduces the average depth per round, and the strategy generalizes naturally to higher-dimensional product codes. This scheduling framework enables computational speedups for quantum computing architectures built on these codes, all while preserving high-fidelity error correction.

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1 Introduction

Fault-tolerant quantum computing (FTQC) relies on quantum error correction (QEC) to suppress hardware noise. Although surface-code QEC has been demonstrated experimentally [1, 8], scaling to large-scale FTQC motivates codes with higher encoding rates to keep space (i.e., qubit) overhead practical. This leads to quantum low-density parity-check (qLDPC) codes [3] with improved encoding rates and low-weight parity checks, which in principle enable shallow syndrome extraction circuits to measure the checks. Because two-qubit gates in these circuits propagate errors between data and ancilla qubits, reducing circuit depth is critical for limiting error accumulation and latency of QEC rounds, thereby improving both fidelity and speed of FTQC.

This work presents a syndrome extraction circuit construction strategy that applies broadly across the main qLDPC code families proposed. In particular, it applies to homological product codes such as hypergraph product (HP) codes [23], lifted-product codes

including quasi-cyclic lifted-product codes [16, 29] and other quasi-abelian variants, as well as certain balanced product codes such as bicycle [2, 13] and tricycle [9, 10, 15] codes. These families share a fundamental algebraic product structure that defines their quantum Tanner graphs. Consequently, our strategy applies uniformly across them, guaranteeing near-optimal depths while respecting non-trivial correctness constraints of syndrome extraction circuits. This generality ensures that the proposed strategy covers essentially all structured qLDPC codes of current practical interest.

A fundamental depth lower bound of the syndrome extraction circuit is the maximum degree in the quantum Tanner graph, Δ , of the code. Contributions of this paper includes:

- A general scheduling strategy for quasi-abelian lifted-product (QALP) codes (including hypergraph product and bivariate bicycle codes) achieving two-qubit depth $\leq \Delta + 1$. For a broad subset of codes, the strategy attains depth Δ , thus optimal.
- The construction naturally categorizes QALP codes into two types based on whether depth Δ is achievable (Type-0) or depth $\Delta + 1$ is necessary (Type-1). An SMT (satisfiability modulo theories) formulation of the scheduling problem is provided to pin-point the type for small-scale codes.
- A pipelined multi-round schedule variant with average depth per round approaching Δ (optimal) with more rounds.
- A generalization to higher-dimensional product codes, e.g., 3D QALP codes [9, 10, 15], that achieves depth within a small additive constant to Δ .
- The scheduling strategy still leaves significant freedom. Numerical simulations with circuit-level noise shows that, although many correct schedules meet the same depth bound, their logical error rates can differ significantly—highlighting the value of the freedom in the scheduling strategy.

The organization of the paper is as follows. Sec. 2 reviews prior work. Sec. 3 introduces the requisite background on quantum error correcting codes and the syndrome extraction circuit construction problem. Sec. 4 presents our scheduling strategy and proves its correctness. Sec. 5 extends the approach to pipelined multi-round schedules and higher-dimensional product codes. Sec. 6 reports numerical results demonstrating the impact of scheduling choices on logical error rates. Sec. 7 concludes with future directions.

2 Related Work

To our knowledge, no prior work provides a unified strategy that, for the broad class of qLDPC codes encompassed by QALP (including HP and generalized bicycle/tricycle families), systematically constructs single-round syndrome extraction circuits whose depth is provably within one layer of the fundamental lower bound.

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A baseline construction for arbitrary CSS codes (and thus all QALP codes) measures X- and Z-type checks in separate steps. Optimal edge coloring of the X- and Z-subgraphs of the quantum Tanner graph then yields depth at most 2Δ ; see Ref. [26].

If an optimal-depth (depth Δ) single-round schedule is known, trivially repeating it for multiple rounds preserves optimal average depth per round. Ref. [29] proposes a pipelined strategy for HP codes that approaches optimal depth over many rounds, but it does not supply a guaranteed single-round construction nor supply a generalization to all QALP codes.

Ref. [24] gives heuristics for HP codes with achieved depths between Δ and 2Δ . Their best case (optimal depth Δ) requires both underlying classical Tanner graphs to have only even-degree nodes and to admit a suitable ‘balanced ordering’; no explicit procedure for finding such an ordering is provided.

Ref. [27] formulates the general CSS scheduling problem as an integer program (IP), enabling optimal solutions for small instances (analogous to our SMT formulation). To scale, they apply the IP locally to subsets of checks, yielding a heuristic without worst-case depth guarantees.

In contrast, our method guarantees, for every QALP code, a single-round schedule of depth $\leq \Delta + 1$, attains the optimal depth Δ on many instances, and extends to (i) multi-round pipelining with average depth approaching Δ and (ii) higher-dimensional lifted products. Prior tailored circuit constructions for specific code families [2, 9, 15, 17, 28] can often be interpreted as special cases from this general scheduling strategy.

3 Background

For a binary linear error-correcting code on n bits, a *parity-check matrix* H is an $m \times n$ binary matrix whose i -th row defines a parity check: the parity of bits in $\{j \mid H_{ij} = 1\}$ must be even. For example, $A = (1, 1, 1)$ is the parity check matrix of the single-parity-check (SPC) code on three bits: A constrains codewords to three-bit strings with even parity. The *Tanner graph* of H , $\mathcal{G}_H$, is a bipartite graph with n bit vertices and m check vertices, and an edge between bit j and check i if and only if $H_{ij} = 1$. The Tanner graph $\mathcal{G}_A$ is shown in Fig. 1a, where solid dots denote bits and squares denote checks.

3.1 Calderbank-Shor-Steane (CSS) Codes

Classical bit measures 0 or 1, whereas the outcome of measuring a qubit depends on the measurement basis. A CSS code is specified by two binary parity-check matrices H_x and H_z that define X- and Z-basis parity checks, respectively. Columns correspond to data qubits that store quantum information, denoted by Q ; rows specify parity checks, denoted by X and Z . For example, check X_i enforces that the X-basis parity of qubits in $\{j \mid (H_x)_{ij} = 1\}$ is even. The Tanner graph of a CSS code $\mathcal{G} = (Q, X \sqcup Z, E)$ is an equivalent representation to H_x and H_z , where $\sqcup$ denotes disjoint union; see Fig. 1a, where delineated squares denote X-checks, diamonds denote Z-checks, and circles denote data qubits.

To ensure that the checks can be measured simultaneously, they must commute, i.e., $H_x H_z^T \equiv 0 \pmod{2}$, i.e., any X_i and Z_j must share an even number of qubits. In the Tanner graph, this means that for any pair (X_i, Z_j) , the size of $N(X_i) \cap N(Z_j)$ is even, where $N(v)$ is the neighborhood of v .

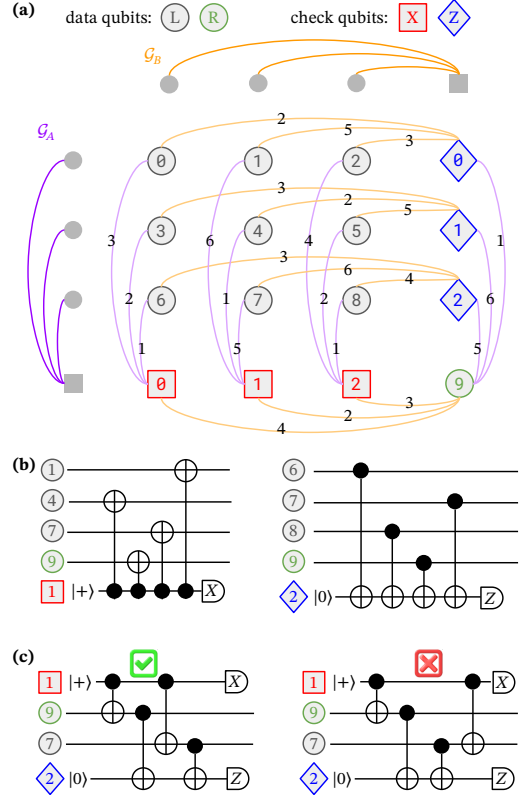


Figure 1: (a) Example hypergraph product code $HP(A, B)$ from parity-check matrix $A = B = (1, 1, 1)$. Shapes with border lines denote data and check qubits; edges denote CNOTs, with labels indicating their scheduled time steps. (b) Syndrome extraction subcircuits for X- and Z-checks; the CNOT order within an individual check may be arbitrary. The $\bullet$ and $\oplus$ symbols denote control and target qubits of CNOTs, respectively. (c) Global scheduling constraint: if an X-check and a Z-check share two data qubits, the ordering of their CNOTs must be consistent—either X-check CNOTs occur before Z-check CNOTs on both data qubits, or both occur after.

We focus on lifted-product codes later on, but first recall hypergraph product codes, a more straightforward special case.

DEFINITION 1 (HYPERGRAPH PRODUCT CODES). Let A and B be binary parity-check matrices of dimensions $m_A \times n_A$ and $m_B \times n_B$, respectively. The hypergraph product code $HP(A, B)$ is the CSS code with

$$H_x = (A \otimes I_{n_B} \quad I_{m_A} \otimes B^T), \quad H_z = (I_{n_A} \otimes B \quad A^T \otimes I_{m_B}). \quad (1)$$

The Tanner graph of $HP(A, B)$ for $A = B = (1, 1, 1)$ is shown in Fig. 1a. Intuitively, it is the Cartesian product of the Tanner graphs of A and B . For HP codes, the data qubits Q can be categorized into two sectors, $Q = \mathcal{L} \sqcup \mathcal{R}$. L-data ($\mathcal{L}$) are the product of A-bits and B-bits, while R-data ($\mathcal{R}$) are the product of A-checks and B-checks. X-checks are the product of A-checks and B-bits, while Z-checks are the product of A-bits and B-checks. Edges are inherited from $\mathcal{G}_A$ and $\mathcal{G}_B$ in the natural way. The product of the maximum-degree vertices in the factor graphs yields the maximum-degree qubit (e.g.,

$Q_9 \in \mathcal{R}$ in Fig. 1a). Thus $\Delta_{\text{HP}(A,B)} := \Delta(\mathcal{G}_{\text{HP}(A,B)}) = \Delta_A + \Delta_B$, where Δ denotes the maximum vertex degree.

3.2 Syndrome Extraction Circuits

A parity check cannot be implemented by measuring each data qubit and classically computing the parity, as that would destroy the quantum superposition. Instead, for Z-checks we initialize an ancilla in $|0\rangle$ (resp. $|+\rangle$ for X-checks), apply CNOTs from data to Z-ancillae (resp. from X-ancillae to data), then measure the ancilla in the Z- (resp. X-) basis; see Fig. 1b. Within a single check, the order of its CNOTs is arbitrary, provided one CNOT per data qubit in the check is applied, and the CNOTs are in the correct direction. However, there are global ordering constraints between the checks. If checks X_i and Z_j share exactly two data qubits q and q' , then their CNOTs must be ordered consistently on q and q' : on both data qubits, either CNOTs from X_i occur before CNOTs to Z_j , or both X_i CNOTs occur after Z_j CNOTs; see Fig. 1c. Edges in the Tanner graph correspond one-to-one with CNOTs: along an edge, the CNOT is from the data qubit to a Z-check or from an X-check to the data qubit.

We focus on the following scheduling problem.

PROBLEM 1 (DEPTH MINIMIZATION FOR SINGLE-ROUND SINGLE-ANCILLA CSS-CODE SYNDROME EXTRACTION). *Given a CSS code with Tanner graph $\mathcal{G} = (Q, X \sqcup Z, E)$, find the minimum CNOT depth D and a scheduling function $t : E \rightarrow \{1, \dots, D\}$ to assign CNOTs to time steps such that (i) no two edges incident on the same vertex share a time step, and (ii) for every pair of checks $X_i \in X$ and $Z_j \in Z$,*

$$0 < \prod_{q \in N(X_i) \cap N(Z_j)} (t_{(X_i, q)} - t_{(q, Z_j)}). \quad (2)$$

Condition (ii) [27] generalizes the consistent ordering when an X-check and a Z-check share more than two data qubits. A trivial lower bound is $D \geq \Delta_{\mathcal{G}}$, since a maximum-degree qubit can participate in at most one CNOT per time step. The scheduling function shown as edge labels in Fig. 1a satisfies these constraints, and it is optimal because $D = 6 = \Delta_{\text{HP}(A,B)}$.

3.3 Lifted-Product Codes

Recall that an $l \times l$ permutation matrix has exactly one entry equal to 1 in each row and column. Two permutation matrices are *non-overlapping* if they never have a 1 in the same position. *Lifting* replaces binary entries with structured $l \times l$ blocks; we bypass the ring-theoretic treatment [16] and work directly with block matrices. For a matrix M , we denote its (i, j) entry by M_{ij} or $(M)_{ij}$ for clarity when M is already a complex expression.

DEFINITION 2 (LIFTED PARITY-CHECK MATRIX). *An l -lifted parity-check matrix $\tilde{H}$ of dimension $m \times n$ is a block matrix whose (i, j) block is $\tilde{H}_{ij}$, a sum of permutations*

$$\tilde{H}_{ij} = \sum_{k=1}^{(W_{\tilde{H}})_{ij}} \tilde{H}_{ij}^{(k)}, \quad (3)$$

where the $\tilde{H}_{ij}^{(k)}$ are pairwise non-overlapping $l \times l$ permutation matrices, and the weight matrix $W_{\tilde{H}} \in \{0, 1, \dots, l\}^{m \times n}$ records the number of permutations of the blocks. Expanding the blocks to binary matrices, $\tilde{H}$ expands to a parity-check matrix H with dimension $ml \times nl$.

The lifted Tanner graph $\mathcal{G}_{\tilde{H}}$ is a bipartite *multigraph*: each lifted vertex $\tilde{v}$ represents a set of l bits (or checks), and each lifted edge $\tilde{e}$ represents l couplings between such sets. The regular Tanner graph $\mathcal{G}_H$ is obtained by expanding each $\tilde{v}$ into l vertices and each $\tilde{e}$ into l edges, wired according to the corresponding permutation matrices $\tilde{H}_{ij}^{(k)}$. By construction, different $\tilde{e}$ expand to different sets of edges in $\mathcal{G}_H$, i.e., there is an injective map $e \mapsto \tilde{e}$. Thus, the degree of v in $\mathcal{G}_H$ equals the degree of the corresponding lifted vertex $\tilde{v}$ in $\mathcal{G}_{\tilde{H}}$ (bit degrees are column sums of $W_{\tilde{H}}$; check degrees are row sums). This means *the maximum degrees of lifted and regular Tanner graphs are the same*, i.e., $\Delta_H := \Delta(\mathcal{G}_H) = \Delta(\mathcal{G}_{\tilde{H}}) =: \Delta_{\tilde{H}}$, so in Δ notation below, it does not matter to keep the tilde or not.

In Fig. 2a we illustrate a 3-lifted parity-check matrix $\tilde{B}$ with weight matrix $W_{\tilde{B}} = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 1 & 2 \end{pmatrix}$ with bits and checks indexed from left to right. In the highlighted region of $\mathcal{G}_{\tilde{B}}$, two lifted edges connect the leftmost bit and check; they expand (see call-out) to two sets of edges in $\mathcal{G}_B$ between a 3-set of bits and a 3-set of checks, arising from two non-overlapping permutations: identity (orange) and a cyclic shift (blue). In contrast, no two lifted edges in $\mathcal{G}_{\tilde{A}}$ connect the same bit and check, so $W_{\tilde{A}} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$ has only entries 0 and 1.

DEFINITION 3 (QUASI-ABELIAN LIFTED-PRODUCT CODES). *Let $\tilde{A}$ and $\tilde{B}$ be l -lifted parity-check matrices of sizes $m_A \times n_A$ and $m_B \times n_B$ with the quasi-abelian requirement: $\tilde{A}_{ij}^{(k)}$ and $\tilde{B}_{i'j'}^{(k')}$ commute for all possible i, j, i', j', k, k' . Let their (expanded) parity-check matrices be A and B . The quasi-abelian lifted-product code $\text{QALP}(\tilde{A}, \tilde{B})$ is the CSS code with*

$$H_X = (A \otimes I_{n_B} \quad I_{m_A} \otimes B), \quad H_Z = (I_{n_A} \otimes B^T \quad A^T \otimes I_{m_B}). \quad (4)$$

As with hypergraph products, the Tanner graph of $\text{QALP}(\tilde{A}, \tilde{B})$ is the Cartesian product of the lifted Tanner graphs $\mathcal{G}_{\tilde{A}}$ and $\mathcal{G}_{\tilde{B}}$, so $\Delta_{\text{QALP}(\tilde{A}, \tilde{B})} = \Delta_{\tilde{A}} + \Delta_{\tilde{B}} (= 2 + 3 = 5$ in Fig. 2a), which lower-bounds the CNOT depth in Problem 1.

4 Three-Stage Scheduling Strategy

We will give a constructive proof of the following result.

THEOREM 1. *For any quasi-abelian lifted-product code $\text{QALP}(\tilde{A}, \tilde{B})$ there exists a single-round single-ancilla (per check) syndrome extraction circuit with CNOT depth at most $\Delta + 1$ where Δ is the maximum degree of the Tanner graph of the code.*

We first expand on near-optimality, then give one valid schedule proving the theorem, and finally summarize scheduling freedom.

4.1 Near-Optimal CNOT Depth

The trivial lower bound on the CNOT depth for a QALP code is Δ , as discussed earlier. Our strategy yields depth Δ (thus optimal) for some codes but yields depth $\Delta + 1$ for others. Two questions arise: (i) Do some QALP codes inherently require depth $> \Delta$? (ii) When depth Δ is achievable, can our construction still be suboptimal? Both answers are yes.

We can solve Problem 1 exactly for small instances using an SMT solver [6]. Note the SMT formulation below works for any CSS code. Introduce integer variables t_{ij}^x for each 1 entry in H_X and t_{ij}^z for each 1 entry in H_Z . For a proposed depth D we assert:

- Time step range: $1 \leq t_{ij}^x \leq D, 1 \leq t_{ij}^z \leq D$.

- Edge conflicts (check side): $t_{ij}^x \neq t_{ij'}^x$ for any i and $j \neq j'$; $t_{ij}^z \neq t_{ij'}^z$ for any i' and $j \neq j'$.
- Edge conflicts (data side): $t_{ij}^x \neq t_{i'j}^x$ for any j and $i \neq i'$; $t_{ij}^z \neq t_{i'j}^z$ for any j and $i \neq i'$; $t_{ij}^x \neq t_{i'j'}^z$ for any i, i' , and j .
- Consistent ordering: for every X-check i and Z-check i' sharing two data qubits j and j' :

$$(t_{ij}^x < t_{i'j}^z \wedge t_{ij'}^x < t_{i'j'}^z) \vee (t_{i'j}^z < t_{ij}^x \wedge t_{i'j'}^z < t_{ij'}^x). \quad (5)$$

If X- and Z-checks share more than two data qubits, the ordering constraint (5) needs to generalize according to the expressions in Problem 1. We begin with D = the maximum qubit degree in H_x and H_z and increment until satisfiable.

Evaluation shows that some codes provably require depth $\Delta + 1$. One example is $\text{HP}(H, H)$ where H corresponds to the $K_{4,3}$ complete bipartite graph. Thus, we classify QALP codes:

- Type-0: optimal depth is Δ .
- Type-1: optimal depth is $\Delta + 1$ (our schedule always optimal).

Empirically, denser parity-check matrices correlate with Type-1 but we so far do not have an efficient characterization of the types. For Type-0 codes our construction may incur a single additional layer; e.g. Fig. 1a shows an optimal depth-6 solution of a Type-0 code while our strategy in this section gives depth 7.

4.2 Constructive Proof

We use the coordinate system in Fig. 2c (left). Each qubit has a triple index: (row, column, lift-position). Ranges are:

- L-data qubits: $(j, j', y) \in [n_A] \times [n_B] \times [L]$,
- X-check qubits: $(i, j', x) \in [m_A] \times [n_B] \times [L]$,
- R-data qubits: $(i, i', w) \in [m_A] \times [m_B] \times [L]$,
- Z-check qubits: $(j, i', z) \in [n_A] \times [m_B] \times [L]$.

The linear indexing in H_x and H_z can be calculated by multiplying the ranges accordingly, e.g., the linear index of L-data qubit (j, j', y) is $j \times n_B \times l + j' \times l + y$. From the construction of the parity-check matrices (4), it can be verified that a CNOT from an X-check (i, j', x) to an L-data qubit must preserve the column j' ; its target is (j, j', y) where $(\tilde{A}_{ij}^{(k)})_{xy} = 1$ for some k . Similarly, a CNOT from an R-data qubit (i, i', w) to a Z-check (j, i', z) satisfies $(\tilde{A}_{ij}^{(k)})_{wz} = 1$ for some k . Analogous statements hold for CNOTs given by $\tilde{B}$. Intuitively, this just means that the edges in the lifted Tanner graph for $\text{QALP}(\tilde{A}, \tilde{B})$ are either vertical (associated with $\mathcal{G}_{\tilde{A}}$) or horizontal (associated with $\mathcal{G}_{\tilde{B}}$); see Fig. 2a.

PROOF OF THEOREM 1. The optimal edge coloring of a bipartite multigraph, using exactly its maximum degree, can be efficiently computed [5]. Denote such edge colorings of $\mathcal{G}_{\tilde{A}}$ and $\mathcal{G}_{\tilde{B}}$ by $f : E(\mathcal{G}_{\tilde{A}}) \rightarrow [\Delta_A]$, and $f' : E(\mathcal{G}_{\tilde{B}}) \rightarrow [\Delta_B]$, respectively; see Fig. 2a left for examples. Recall that each of the edges corresponds uniquely to a permutation matrix $\tilde{A}_{ij}^{(k)}$ or $\tilde{B}_{i'j'}^{(k')}$ in the lifted parity-check matrices. Partition CNOTs (i.e., edges) by the colors:

- $E_{x,c}$: all CNOTs from X-check (i, j', x) to L-data (j, j', y) for all $j' \in [n_B]$ with $(\tilde{A}_{ij}^{(k)})_{xy} = 1$ and $f(\tilde{A}_{ij}^{(k)}) = c$.
- $E_{z,c}$: all CNOTs from R-data (i, i', w) to Z-check (j, i', z) for all $i' \in [m_B]$ with $(\tilde{A}_{ij}^{(k)})_{wz} = 1$ and $f(\tilde{A}_{ij}^{(k)}) = c$.
- $E'_{x,c'}$ and $E'_{z,c'}$ defined analogously for $\tilde{B}$.

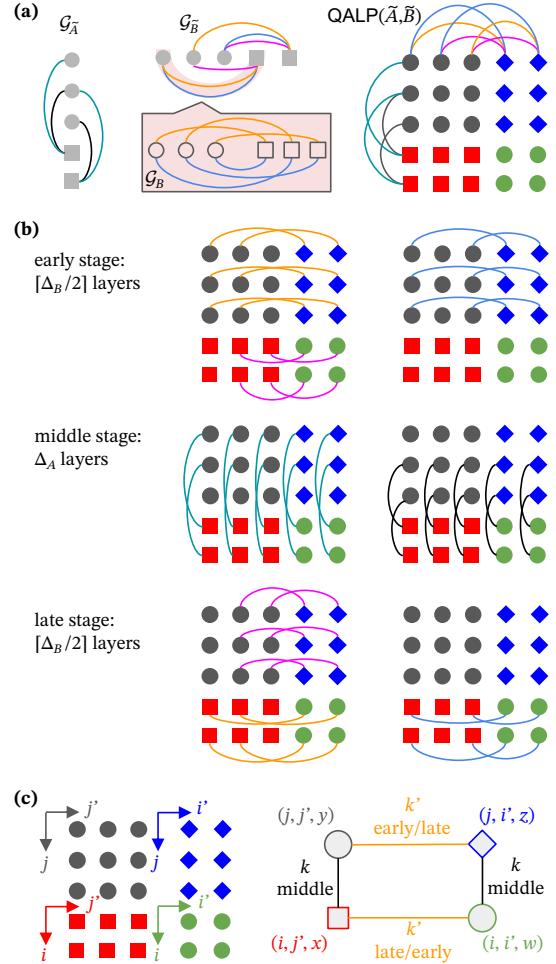


Figure 2: (a) Edges in the lifted Tanner multigraph correspond to individual permutations in the lifted parity-check matrix. The call-out shows the highlighted part in the corresponding regular Tanner graph. Optimal edge coloring uses Δ colors. (b) Three-stage schedule: early stage uses first half of $\tilde{B}$'s colors on Z (orange and blue edges) and the complementary colors on X (magenta edges), middle stage uses all $\tilde{A}$ colors, late stage uses complementary colors to the early stage (magenta on Z, orange and blue on X). (c) Left: qubit coordinate system. Right: correctness proof sketch—shared data qubits between any X-/Z-check form L/R pairs; horizontal CNOTs for each pair are separated across early/late stages, enforcing consistent ordering.

Schedule the CNOTs in three stages:

- (1) Early stage (half the colors of $\tilde{B}$): for $s = 1, \dots, \lceil \Delta_B/2 \rceil$, apply all CNOTs in $E'_{z,s}$ and $E'_{x,s+\lceil \Delta_B/2 \rceil}$ (note the latter set is empty by definition if $s + \lceil \Delta_B/2 \rceil > \Delta_B$).
- (2) Middle stage (all colors of $\tilde{A}$): for $s = 1, \dots, \Delta_A$ apply all CNOTs in $E_{z,s}$ and $E_{x,s}$.
- (3) Late stage (remaining colors of $\tilde{B}$): for $s = 1, \dots, \lceil \Delta_B/2 \rceil$, apply $E'_{x,s}$ and $E'_{z,s+\lceil \Delta_B/2 \rceil}$.

The schedule is visualized as the circuit in Fig. 3a where each CNOT symbol represents a group of CNOTs. In the early stage,

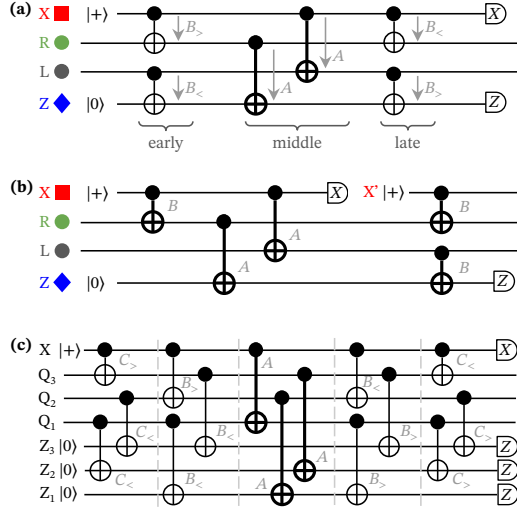


Figure 3: (a) Three-stage schedule for QALP codes. Each drawn CNOT represents a group of physical CNOTs; thick symbols apply all colors of $\tilde{A}$, while thin symbols apply half of the colors of $\tilde{B}$, denoted $B_{<}$ (first half) and $B_{>}$ (second half). (b) Pipelined multi-round schedule: half of the X-check gates in the next round overlap with half of the Z-check gates in the current round. (c) Five-stage schedule for 3D QALP codes.

$E'_{z, 1 \leq s \leq \lceil \Delta_B/2 \rceil}$, i.e., the CNOTs from L-data to Z-checks associated with the first half of colors in $\tilde{B}$ are implemented, along with $E'_{x, \lceil \Delta_B/2 \rceil < s \leq \Delta_B}$, i.e., the CNOTs from X-checks to R-data associated with the second half of colors in $\tilde{B}$. The schedule is also visualized spatially in Fig. 2b: early stage realizes the first half of $\tilde{B}$ colors on Z-checks (orange and blue edges) and the complementary colors on X-checks (magenta edges).

It is straightforward to verify that no vertex conflict occurs by construction of the CNOT partitions. The total depth is $\Delta_A + \Delta_B$ if Δ_B is even and $\Delta_A + \Delta_B + 1$ otherwise.

It remains to check the global ordering constraints between checks. We can gain some intuition with Fig. 3a. For a pair of X-check and Z-check, consider the path from the X-check through data qubits to the Z-check: it can be from X-check to R-data via $B_{>}$ (early stage), and then from R-data to Z-check via A (middle stage); or from X-check to L-data via A (middle stage), and then from L-data to Z-check via $B_{>}$ (late stage). The quasi-abelian condition of the code ensures that the two types of paths are in one-to-one correspondence, so the shared data qubits between the X-check and Z-check form L/R pairs. One can verify that the four CNOTs on these two paths in Fig. 3a follows a correct pattern like Fig. 1c.

Formally, fix an X-check (i, j', x) and a Z-check (j, i', z) . Since these commute, they share an even number of data qubits. We build a bijection between shared L-data and shared R-data qubits. A shared L-data qubit has coordinates (j, j', y) with $(\tilde{A}_{ij}^{(k)})_{xy} = 1$ and $(\tilde{B}_{i'j'}^{(k')})_{yz} = 1$ for some k and k' . Since these are permutation matrices, it holds that $(\tilde{A}_{ij}^{(k)} \tilde{B}_{i'j'}^{(k')})_{xz} = 1$. Then, the quasi-abelian condition in Definition 3 implies $(\tilde{B}_{i'j'}^{(k')} \tilde{A}_{ij}^{(k)})_{xz} = 1$. Since these are two permutation matrices, there is a unique w with $(\tilde{B}_{i'j'}^{(k')})_{xw} = 1$

and $(\tilde{A}_{ij}^{(k)})_{wz} = 1$, giving a shared R-data qubit (i, i', w) . Hence, we proved there is an injective map from the shared L-data to the shared R-data qubits, and then by symmetry, one can show this map is bijective. Thus shared qubits form L/R pairs.

Within each pair (see Fig. 2c right) vertical CNOTs (those involving $\tilde{A}$) occur in the middle stage; the two horizontal CNOTs (involving $\tilde{B}$) are separated—one scheduled early, the other late—because they originate from the same lifted edge color and are placed in complementary halves. Hence the relative order of the two CNOTs attached to the X-check vs. those attached to the Z-check is consistent across both qubits in the pair, giving a positive product. Multiplying over all pairs preserves positivity, and thus satisfies condition (ii) of Problem 1. Therefore, the schedule is correct and achieves depth at most $\Delta_A + \Delta_B + 1 = \Delta + 1$. $\square$

4.3 Freedom in Scheduling

The essence of the correctness proof is that all CNOTs in one direction are placed in the middle stage, while the CNOTs in the other direction are split into two complementary halves executed before and after the middle stage. This pattern leaves substantial freedom, summarized below. It yields many schedules with the target depth, enabling further selection based on noise performance.

- *Choice of middle stage:* either the vertical edges or the horizontal edges can occupy the middle stage; edges in the other direction are split between the early and late stages.
- *Edge coloring:* any optimal edge coloring of the lifted Tanner graphs of $\tilde{A}$ and $\tilde{B}$ may be used.
- *Early/late assignment:* colors have no inherent order, so any half of the colors can be designated as early, with the complementary half as late.
- *In-stage ordering:* within each stage, the order of color layers can be permuted arbitrarily.
- *Color alignment:* within a layer, different rows/columns may use different colors; e.g., in Fig. 2b (middle stage), vertical teal and black edges can be interleaved across columns.

5 Generalizations

We discuss two extensions of our scheduling strategy: (i) pipelined circuits for multiple rounds of syndrome extraction, which may further reduce the average CNOT depth per round, and (ii) a generalization to products of more than two classical codes. Due to space constraints and the similarity of the arguments, we omit full proofs and instead outline the constructions and intuition.

5.1 Pipelined Circuit for Multiple Rounds

As stated in Problem 1, our focus so far has been single-round syndrome extraction, which is broadly useful and especially applicable to single-shot error correction [18]. Additional optimization is possible for multiple rounds. Fig. 3b illustrates an example: the CNOTs associated with B are not split anymore, and the last CNOT group in the current round (from L-data to Z-checks) is executed concurrently with the first CNOT group in the next round (from X-checks to R-data), except in the final round. The resulting CNOT ordering still matches a correct pattern similar to Fig. 1c.

For r rounds, the total CNOT depth is $\Delta_B + r\Delta$, so the average depth per round approaches the single-round lower bound Δ as

r grows. A caveat is that X-check measurement and reset occur mid-schedule; for superconducting qubits, slower measurement compared to other gates may favor aligned per-round measurements. In architectures with reconfigurable connectivity and ample ancilla qubits (e.g., neutral atoms), the pipeline can be practical for the additional speedup. This idea was proposed for HP codes in Ref. [29]; our construction extends it to QALP codes, including generalized bicycle codes.

5.2 Higher-Dimensional Lifted-Product Codes

Definition 3 considers products of two classical codes, but lifted products extend naturally to higher dimensions, e.g., a product of three codes with parity-check matrices [10]:

$$\begin{aligned} H_x &= (A \otimes I \otimes I \quad I \otimes B \otimes I \quad I \otimes I \otimes C), \\ H_z &= \begin{pmatrix} I \otimes B^T \otimes I & A^T \otimes I \otimes I & 0 \\ I \otimes I \otimes C^T & 0 & A^T \otimes I \otimes I \\ 0 & I \otimes I \otimes C^T & I \otimes B^T \otimes I \end{pmatrix}, \end{aligned} \quad (6)$$

where A , B , and C are the expanded parity-check matrices of three lifted parity-check matrices $\tilde{A}$, $\tilde{B}$, and $\tilde{C}$ satisfying the quasi-abelian condition pairwise. The qubits are divided into 3 data sectors, 3 Z-check sectors and one X-check sector. For such a 3D QALP code, our strategy is extended to five stages, as illustrated in Fig. 3c. The middle three stages are similar to the three-stage schedule for 2D QALP codes, and the first and last stages implement the CNOTs associated with C , split into two halves. The CNOT depth for such a five-stage schedule is at most $\Delta_A + \Delta_B + \Delta_C + 2 = \Delta + 2$ since the splitting of $\tilde{B}$ and $\tilde{C}$ may each introduce an extra layer. To check the correctness, one can focus on X-checks and Z-checks from a specific Z-sector, and verify that the shared data qubits again form L/R pairs, and the CNOT ordering on each pair follows a correct pattern. More details can be found in Ref. [15].

6 Numerical Results

Because the synthesized circuits are Clifford circuits, correctness can be checked efficiently via stabilizer simulation. We construct circuits in Stim [7], specify the logical operators of the code, and command Stim to compile a ‘detector error model’, DEM. If successful, this verifies that the intended stabilizers are measured and that logical operators are preserved. The circuit depths can also be simply counted. We implemented a range of QALP codes, including those in Ref. [29]. The circuit depths are as intended and Stim managed to compile DEM successfully for all these codes, which already confirms our main claim, i.e., the constructed syndrome extraction circuits are correct with the target CNOT depth.

Additionally, we explore how the scheduling freedom discussed in Sec. 4.3 affects performance for tricycle codes (a class of 3D QALP codes) under circuit-level noise. The study focuses on the $[[48, 6, (d_z = 4, d_x = 8)]]$ tricycle code from Ref. [15], simulated for $d_z = 4$ rounds of repeated syndrome extraction.

We generated 1,728 candidate schedules induced by the ordering options. For each schedule, we simulate repeated rounds under a two-qubit depolarizing noise model, where each CNOT is followed, with probability p_{2q} , by a random two-qubit Pauli error. This error rate is the characteristic physical error rate for the simulations. Spacetime syndromes are decoded with belief propagation

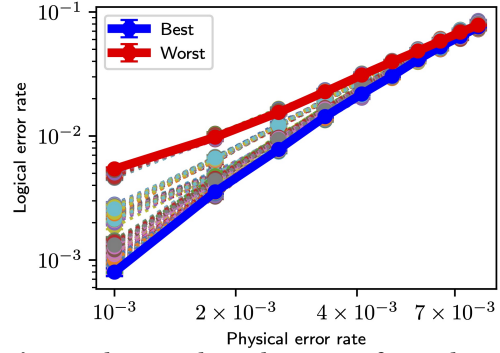


Figure 4: Logical versus physical error rate for syndrome extraction circuits of the $[[48, 6, (d_z = 4, d_x = 8)]]$ tricycle code. Each curve represents a different valid schedule. The data for the best and worst schedules are thickened.

augmented by ordered statistics decoding. Logical error rates are estimated from at least 10^4 Monte Carlo samples per schedule.

As displayed in Fig. 4, logical error rates vary substantially across correct schedules with the same depth—by up to an order of magnitude at fixed physical error rate. This indicates that ordering choices can materially affect fault-tolerant performance for this code.

7 Conclusion and Outlook

In this work, we presented a three-stage schedule for single-round single-ancilla syndrome extraction circuit of quasi-abelian lifted-product codes, encompassing hypergraph product and bicycle/tricycle families. The strategy achieves CNOT depth at most $\Delta+1$, where Δ is the maximum degree of the Tanner graph, a fundamental depth lower bound. With a pipelined extension, the average per-round depth approaches Δ . The construction further extends to higher-dimensional product codes with bounded overhead from optimal depth. Numerical simulations confirm correctness and show that valid schedule choices can markedly affect logical error rates.

Several directions remain open. First, precisely characterizing when depth Δ is attainable (Type-0) versus provably requiring $\Delta+1$ (Type-1) would sharpen the theory and guide code design. Second, extending the construction beyond the quasi-abelian setting is an intriguing open problem. Third, exploiting the scheduling freedom for fault tolerance [4] could yield additional gains: for hypergraph product codes any ordering achieves the code distance [14], whereas for some bivariate bicycle codes no known circuit achieves full code distance [2], making our solutions a useful search space amenable to recent optimization methods [12, 25]. Fourth, customizing the ordering to architecture-specific features—e.g., qubit addressing and rearrangement in neutral-atom systems [19–22], frequency allocation and crosstalk constraints in superconducting platforms [11]—will strengthen the practicality of the results.

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